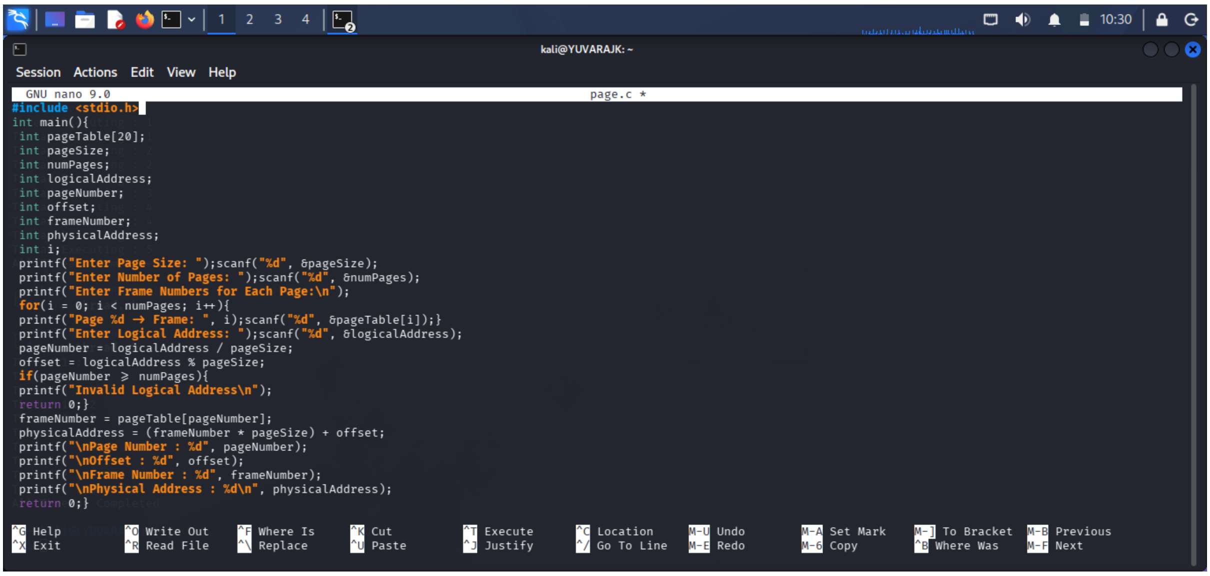
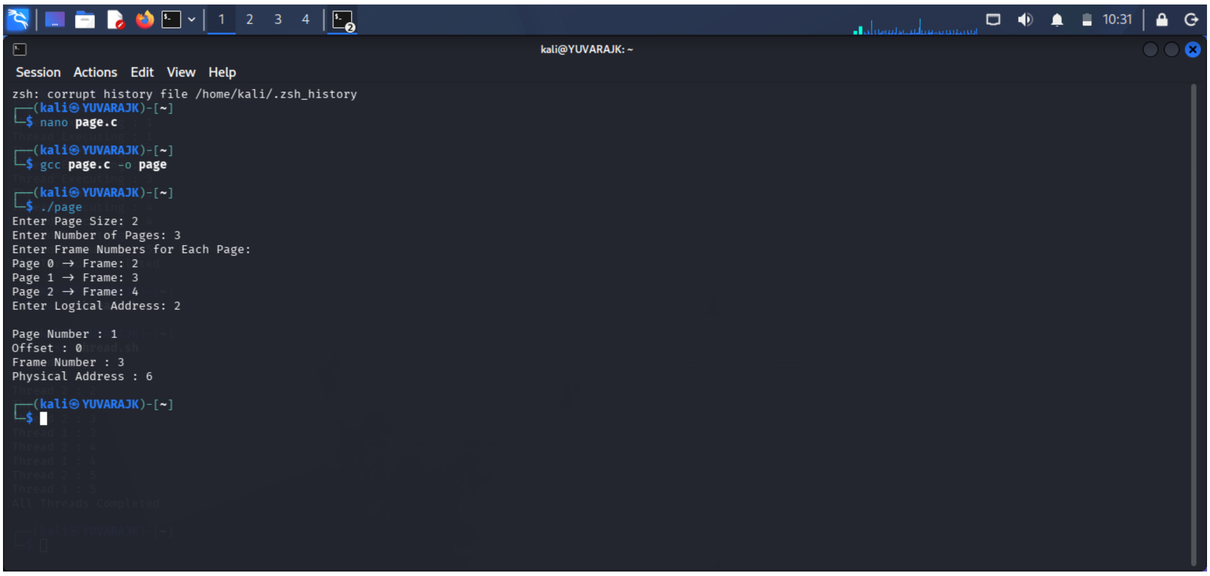
C PROGRAMMING :PAGING



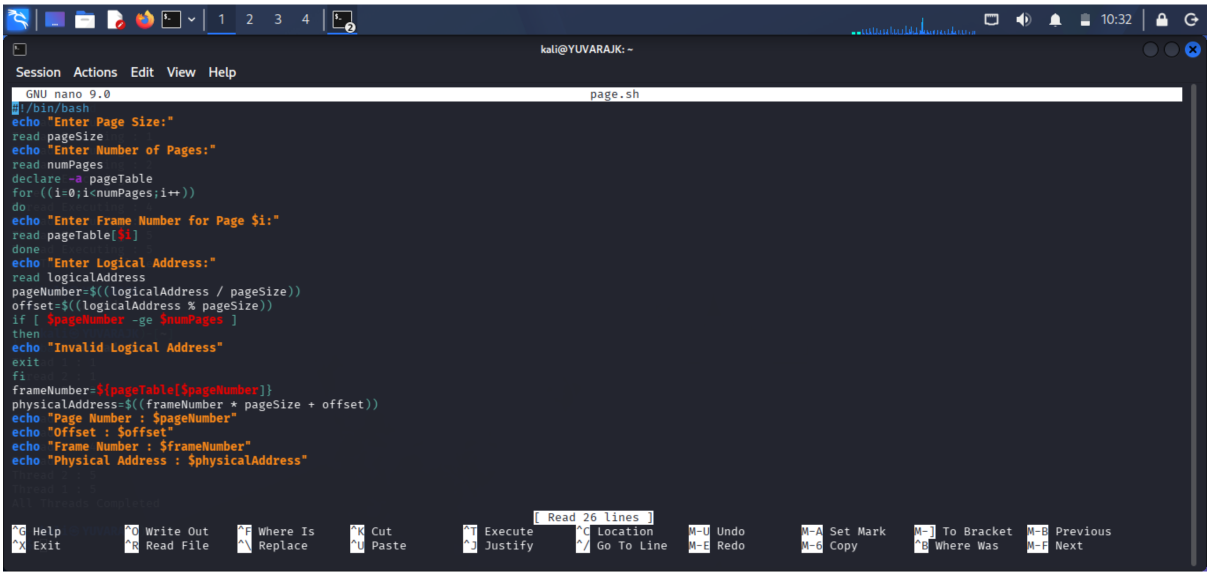
```c
#include <stdio.h>
int main(){
 int pageTable[20];
 int pageSize;
 int numPages;
 int logicalAddress;
 int pageNumber;
 int offset;
 int frameNumber;
 int physicalAddress;
 int i;
 printf("Enter Page Size: ");scanf("%d", &pageSize);
 printf("Enter Number of Pages: ");scanf("%d", &numPages);
 printf("Enter Frame Numbers for Each Page:\n");
 for(i = 0; i < numPages; i++){
 printf("Page %d -> Frame: ", i);scanf("%d", &pageTable[i]);}
 printf("Enter Logical Address: ");scanf("%d", &logicalAddress);
 pageNumber = logicalAddress / pageSize;
 offset = logicalAddress % pageSize;
 if(pageNumber >= numPages){
 printf("Invalid Logical Address\n");
 return 0;}
 frameNumber = pageTable[pageNumber];
 physicalAddress = (frameNumber * pageSize) + offset;
 printf("\nPage Number : %d", pageNumber);
 printf("\nOffset : %d", offset);
 printf("\nFrame Number : %d", frameNumber);
 printf("\nPhysical Address : %d\n", physicalAddress);
 return 0;}
```

OUTPUT :



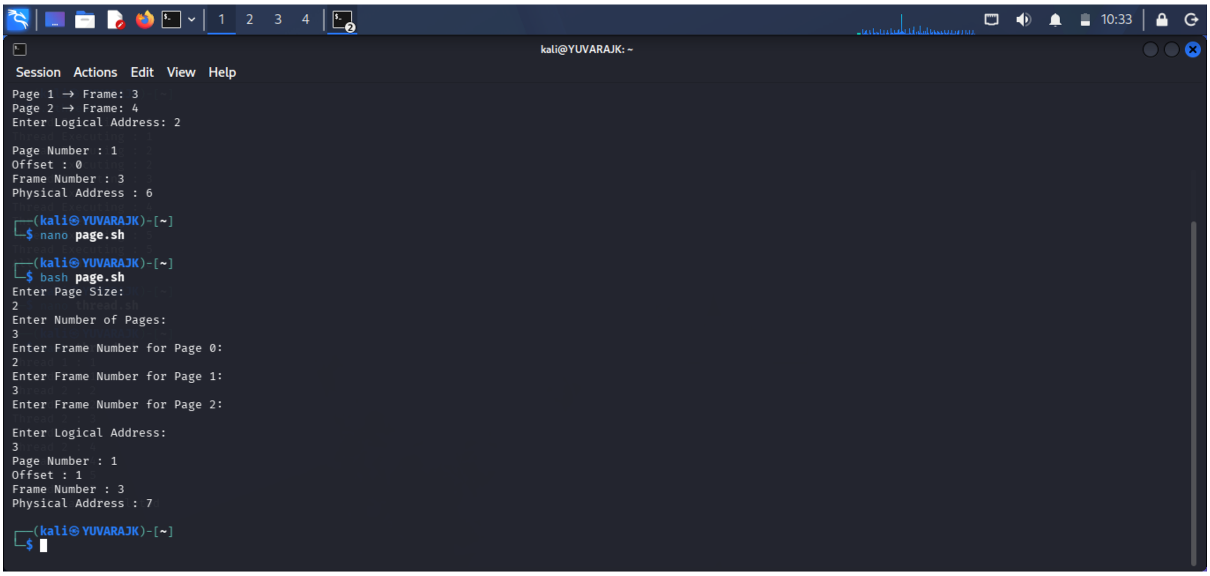
```
zsh: corrupt history file /home/kali/.zsh_history
┌──(kali㉿YUVARAJK)-[~]
└─$ nano page.c

┌──(kali㉿YUVARAJK)-[~]
└─$ gcc page.c -o page

┌──(kali㉿YUVARAJK)-[~]
└─$ ./page
Enter Page Size: 2
Enter Number of Pages: 3
Enter Frame Numbers for Each Page:
Page 0 → Frame: 2
Page 1 → Frame: 3
Page 2 → Frame: 4
Enter Logical Address: 2

Page Number : 1
Offset : 0
Frame Number : 3
Physical Address : 6

┌──(kali㉿YUVARAJK)-[~]
└─$
```

SHELL PROGRAMMING : PAGING



```
GNU nano 9.0                                   page.sh
#!/bin/bash
echo "Enter Page Size:"
read pageSize
echo "Enter Number of Pages:"
read numPages
declare -a pageTable
for ((i=0;i<numPages;i++))
do
echo "Enter Frame Number for Page $i:"
read pageTable[$i]
done
echo "Enter Logical Address:"
read logicalAddress
pageNumber=$((logicalAddress / pageSize))
offset=$((logicalAddress % pageSize))
if [ $pageNumber -ge $numPages ]
then
echo "Invalid Logical Address"
exit
fi
frameNumber=${pageTable[$pageNumber]}
physicalAddress=$((frameNumber * pageSize + offset))
echo "Page Number : $pageNumber"
echo "Offset : $offset"
echo "Frame Number : $frameNumber"
echo "Physical Address : $physicalAddress"
```

OUTPUT :



```
Page 1 → Frame: 3
Page 2 → Frame: 4
Enter Logical Address: 2

Page Number : 1
Offset : 0
Frame Number : 3
Physical Address : 6

┌──(kali㉿YUVARAJK)-[~]
└─$ nano page.sh

┌──(kali㉿YUVARAJK)-[~]
└─$ bash page.sh
Enter Page Size:
2
Enter Number of Pages:
3
Enter Frame Number for Page 0:
2
Enter Frame Number for Page 1:
3
Enter Frame Number for Page 2:

Enter Logical Address:
3
Page Number : 1
Offset : 1
Frame Number : 3
Physical Address : 7

┌──(kali㉿YUVARAJK)-[~]
└─$
```